

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$P = \begin{pmatrix} P_{(+++)} & & & & 0 \\ & P_{(+ - +)} & & & \\ & & P_{(+ - -)} & & \\ 0 & & & \ddots & \\ & & & & P_{(---)} \end{pmatrix}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix}_q = \begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix}_q = (-)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}_{q^{-1}}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & \sum_{i=0}^{m-1} \{ \sum_{i'=0}^{l_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{i'}(g(\sigma_B^i(h(x)))))) - \sum_{j'=0}^{k_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{j'}(g(\sigma_B^i(h(x)))))) \} \\ &= \sum_{i'=0}^{l_2^m(h(x))-1} f(\sigma_C^{i'}(gh(x))) - \sum_{j'=0}^{k_2^m(h(x))-1} f(\sigma_C^{j'}(g(\sigma_B^m(h(x))))). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & -\left(1 - \frac{2M}{r}\right)^{-1} \partial_t^2 \frac{f}{r} + \left(1 - \frac{2M}{r}\right)^{-1} r^{-2} \partial_{r^*} \left(\left(1 - \frac{2M}{r}\right) r^2 \left(1 - \frac{2M}{r}\right)^{-1} \partial_{r^*} \frac{f}{r} \right) \\ & - \frac{l(l+1)}{r^2} \frac{f}{r} - m^2 \frac{f}{r} = 0, \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sigma_B^{k_1(x)}(h(\sigma_A(x))) &= \sigma_B^{l_1(x)}(h(x)), x \in X_A, \\ \sigma_A^{k_2(y)}(h^{-1}(\sigma_B(y))) &= \sigma_A^{l_2(y)}(h^{-1}(y)), y \in X_B, \end{aligned}$$

and

$$\begin{aligned}l_1(x) - k_1(x) &= 1 + b_1(x) - b_1(\sigma_A(x)), x \in X_A, \\l_2(y) - k_2(y) &= 1 + b_2(y) - b_2(\sigma_B(y)), y \in X_B.\end{aligned}$$